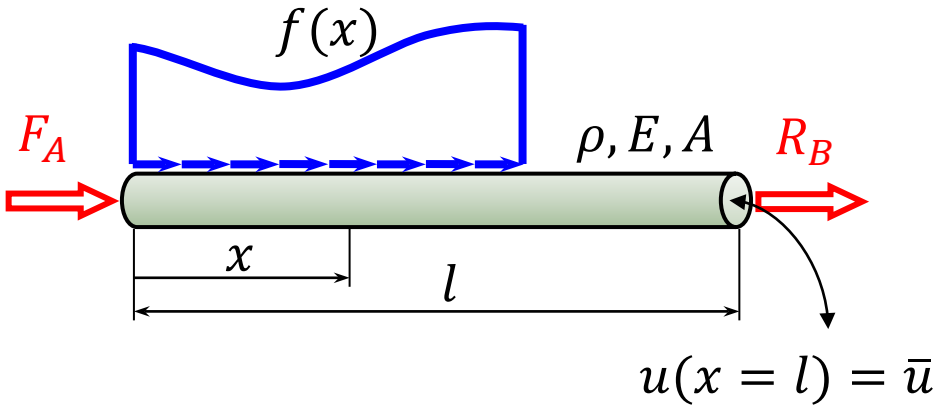
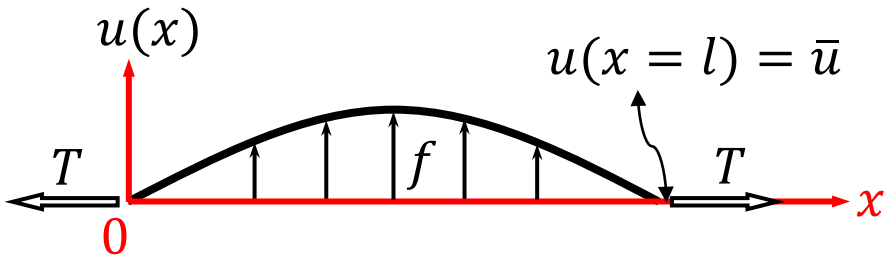
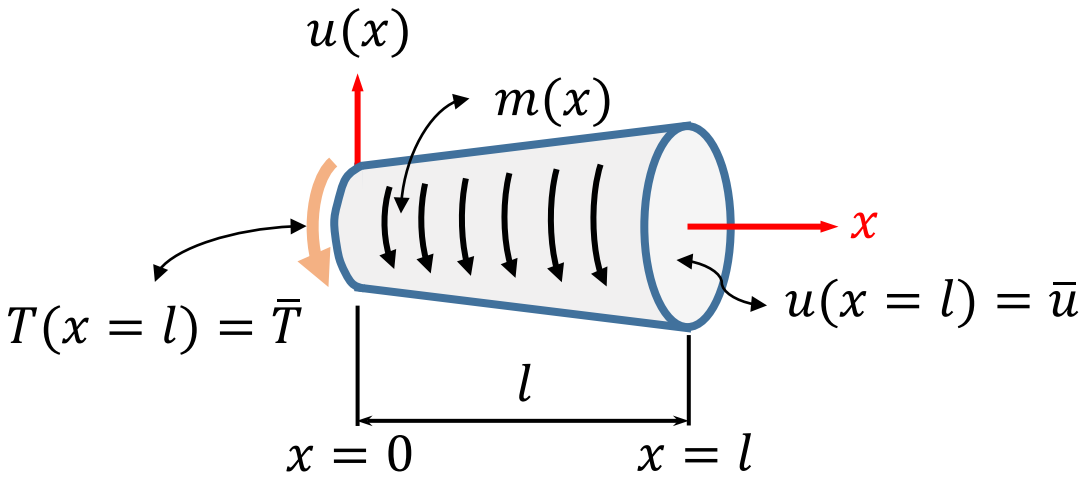
FINITE ELEMENT METHOD
1D PROBLEM

Motivation

Suppose we are interested in solving any of the following problem:







Find $u(x)$ such that

$$\frac{\partial^2 u}{\partial x^2} + f(x) = 0 \qquad 0 < x < l$$

$$u(l) = \bar{u}$$

$$\frac{\partial u}{\partial x}(0) = \bar{v}$$

FEM: Weak form formulation

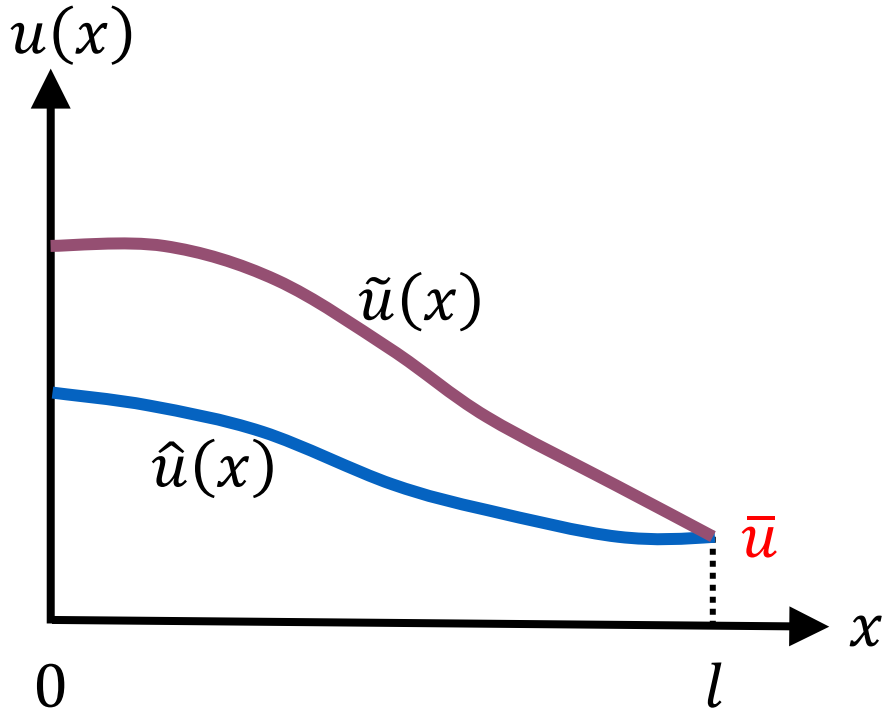
The main idea is to transform the system of differential equations into a set of integral equations.

Partial differential equations with boundary conditions (Boundary value problems)

- Virtual work form
- Weak form
- Integral form
- Variational form

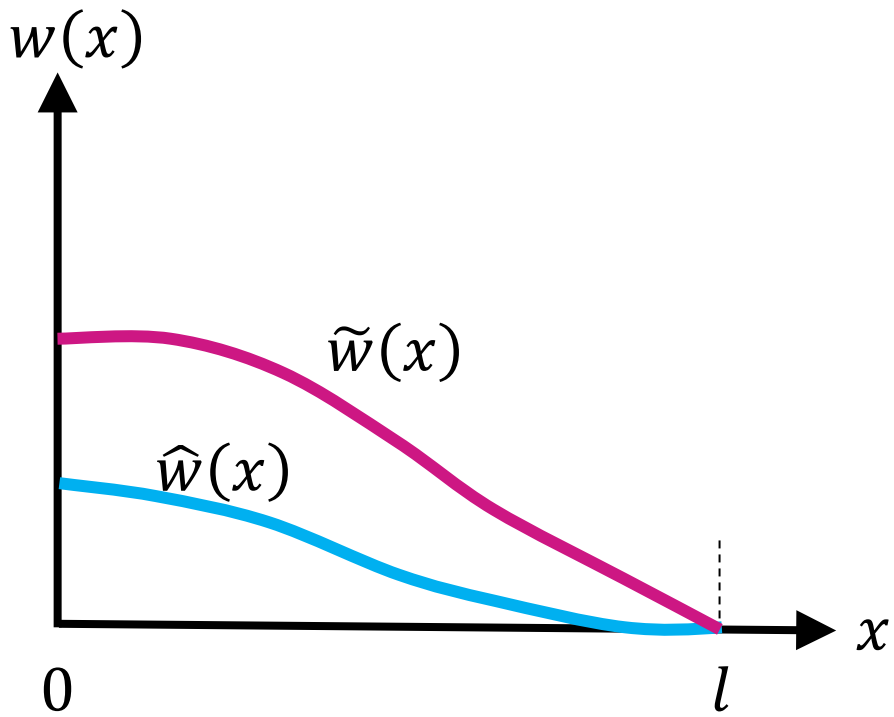
Space functions

Trial functions



$$\mathcal{U} = \{u: [0: l] \to \mathbb{R}, u(l) = \bar{u}\}$$

Test functions



$$\mathcal{W} = \{w: [0: l] \to \mathbb{R}, w(l) = 0\}$$

Procedure for obtaining the weak form

1) Multiply both sides of the equation by a fixed (but arbitrary) test function

$$w(x)\frac{\partial^2 u}{\partial x^2} + w(x)f(x) = w(x)\cdot 0 \qquad \forall w \in \mathcal{W}$$

2) Multiply by the differential length, dx, to obtain the *virtual work of the segment*

$$w(x)\frac{\partial^2 u}{\partial x^2}dx + w(x)f(x)dx = 0\cdot dx \qquad \forall w \in \mathcal{W}$$

3) Integrate, these quantities, along the whole domain

$$\int_0^l w(x)\frac{\partial^2 u}{\partial x^2}dx + \int_0^l w(x)f(x)dx = 0 \qquad \forall w \in \mathcal{W}$$

4) Use integration by parts to reduce the order of the derivative of the first term

$$\int_0^l w\frac{\partial^2 u}{\partial x^2}dx = \left(w\frac{\partial u}{\partial x}\right)_{x=l} - \left(w\frac{\partial u}{\partial x}\right)_{x=0} - \int_0^l \frac{\partial w}{\partial x}\frac{\partial u}{\partial x}dx$$

5) Use the boundary conditions to reduce the problem

$$w(l)\frac{\partial u}{\partial x}(l) = 0 \qquad w(0)\frac{\partial u}{\partial x}(0) = w(0)\bar{v}$$

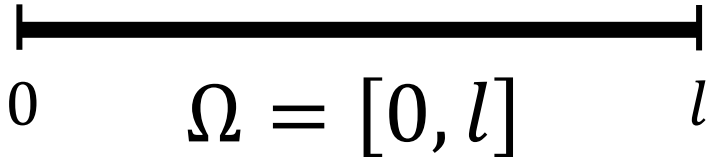
Finally, the weak form of the original differential equations, reads:

Find $u(x)$ such that:

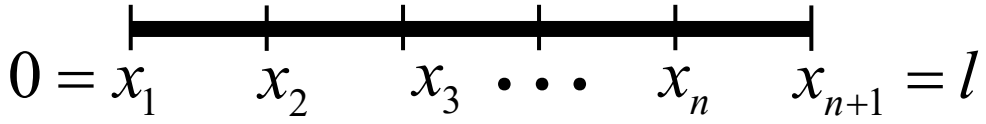
$$\int_0^l \frac{\partial w}{\partial x}\frac{\partial u}{\partial x}dx = -w(0)\bar{v} + \int_0^l wf dx \qquad \forall w \ / \ w(l) = 0$$

Domain approximation

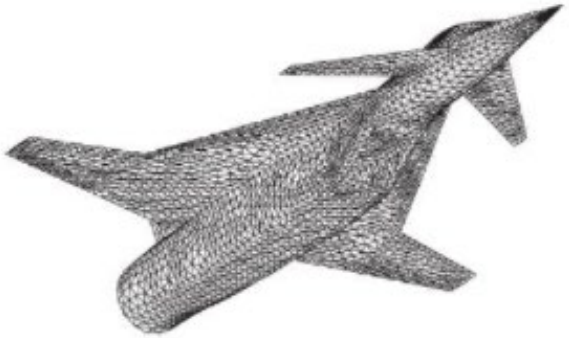
Continuous body

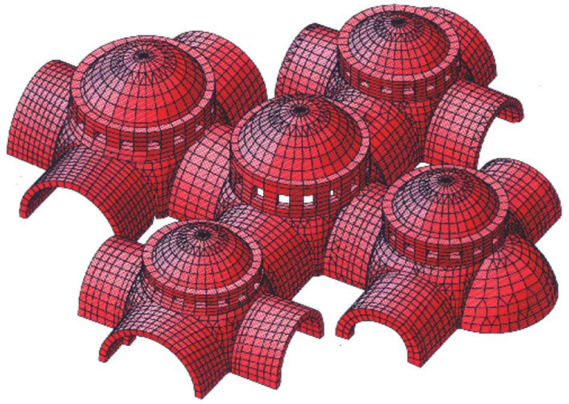


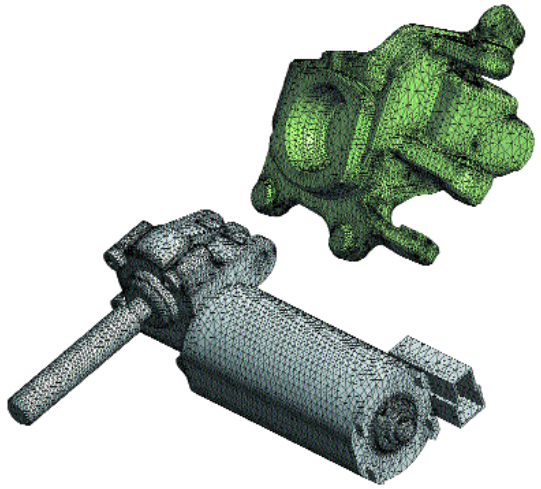
Discrete body

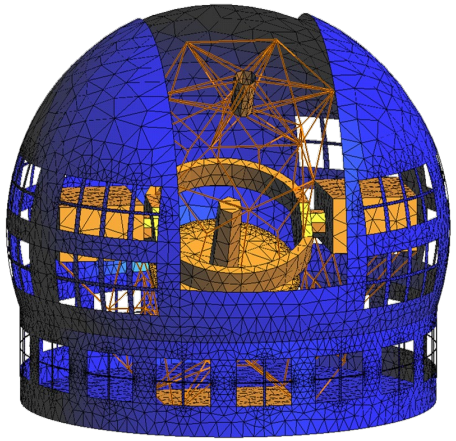


$$\Omega = \bigcup_{A=1}^{n} [x_A, x_{A+1}] = \bigcup_{A=1}^{n} \Omega_A$$



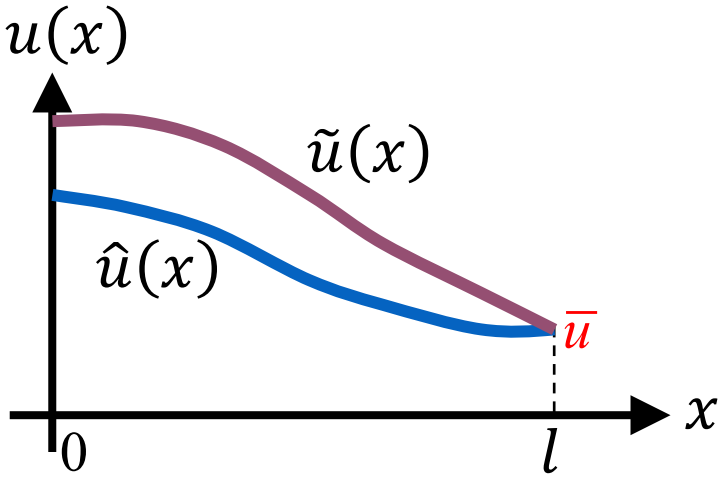


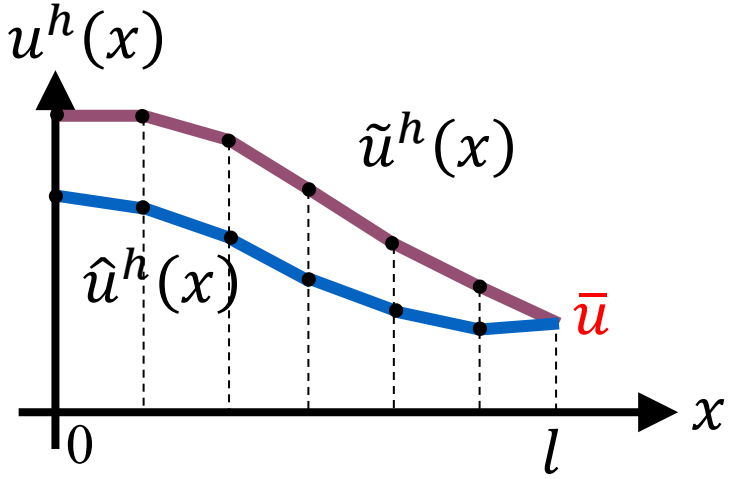




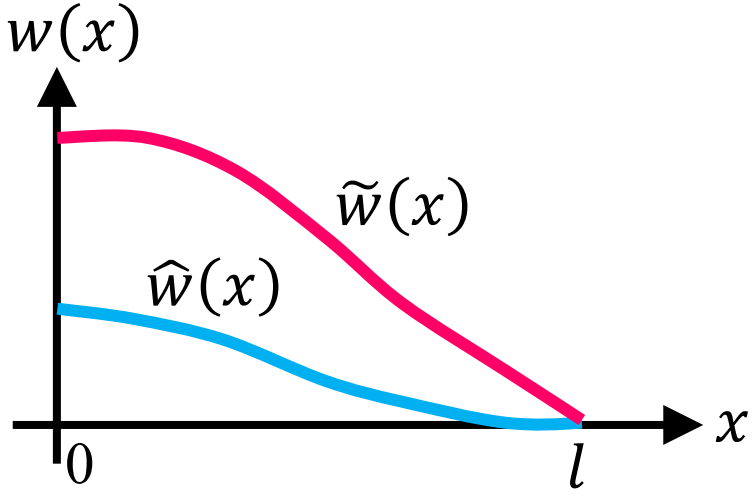
Discretization step

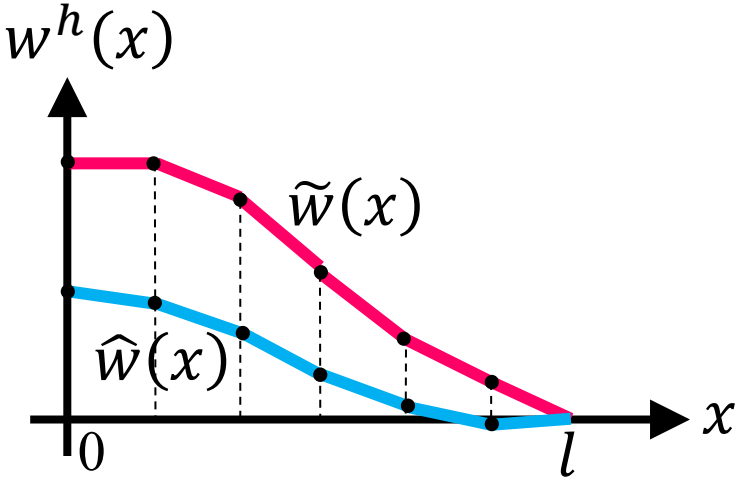
Trial functions





Test functions





Space approximation: Shape functions

$$\mathcal{U} = \{u: [0: l] \to \mathbb{R}, u(l) = \bar{u}\}$$

$$\cong$$

$$\mathcal{U}^h = \{u^h: [0: l] \to \mathbb{R}, u^h(l) = \bar{u}\}$$

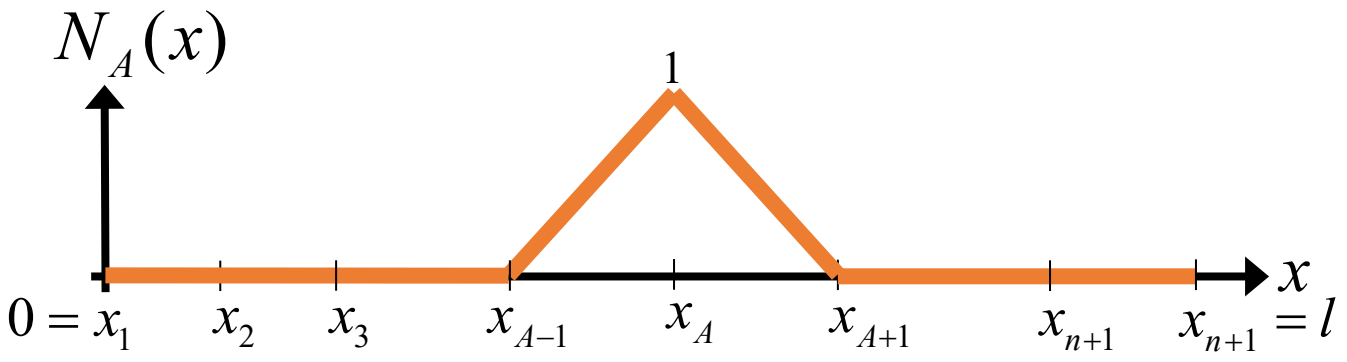
Construct a sequence of finite-dimensional subspaces that approximate $\mathcal{U}$

$$u(x) \cong u^h(x) = \sum_{A=1}^{n+1} u_A N_A(x)$$

$\{N_A(x)\}_{A=1}^{n+1}$ represents the set of basis functions that span the space $\mathcal{U}^h$

For $2 \le A \le n$ internal nodes

$$N_A(x) = \begin{cases} \dfrac{1}{h_{A-1}}\left(x - x_{A-1}\right) & x_{A-1} \le x \le x_A \\ \dfrac{1}{h_A}\left(x_{A+1} - x\right) & x_A \le x \le x_{A+1} \\ 0 & \text{elsewhere} \end{cases}$$

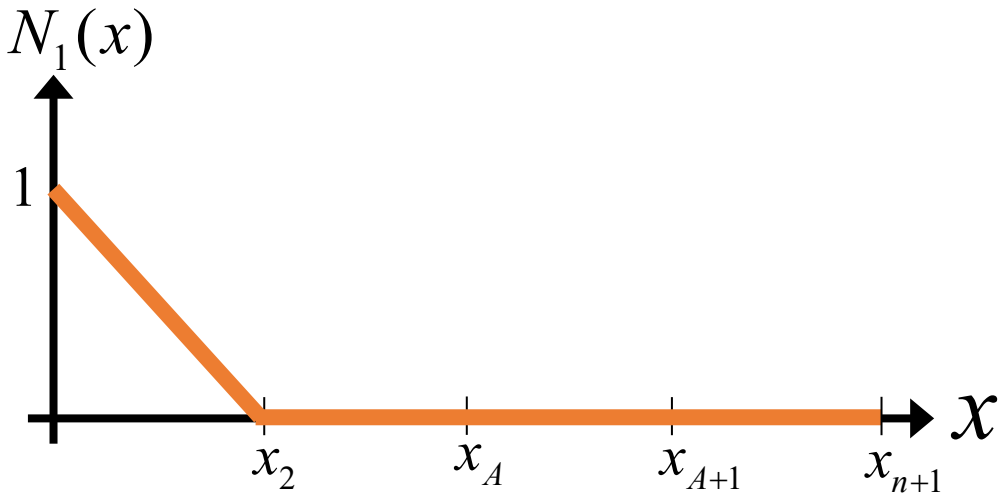


Continuous piecewise linear functions

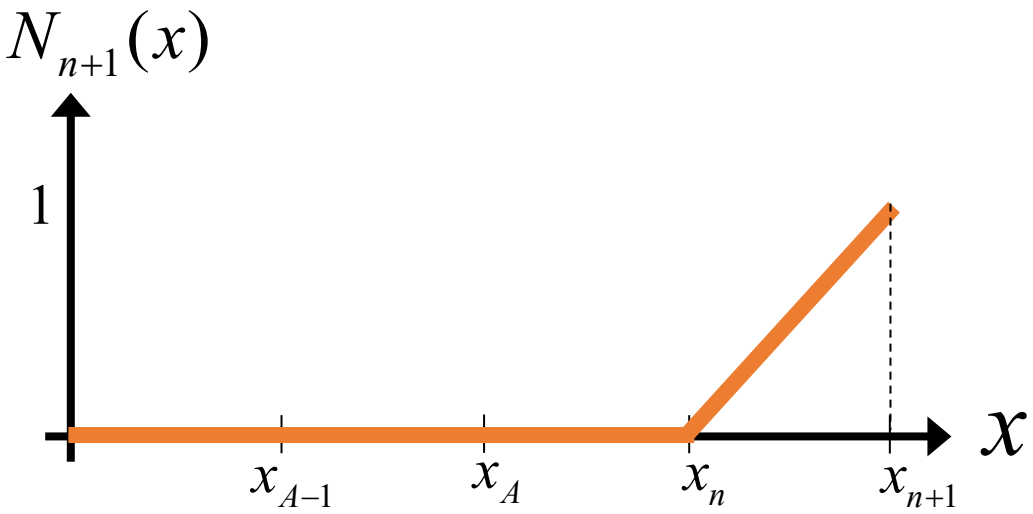
Shape functions at the boundary nodes

For $A = 1,$ and $A = n + 1$ boundary nodes

$$N_1(x) = \frac{x_2 - x}{h_1} \qquad x_1 \le x \le x_2$$

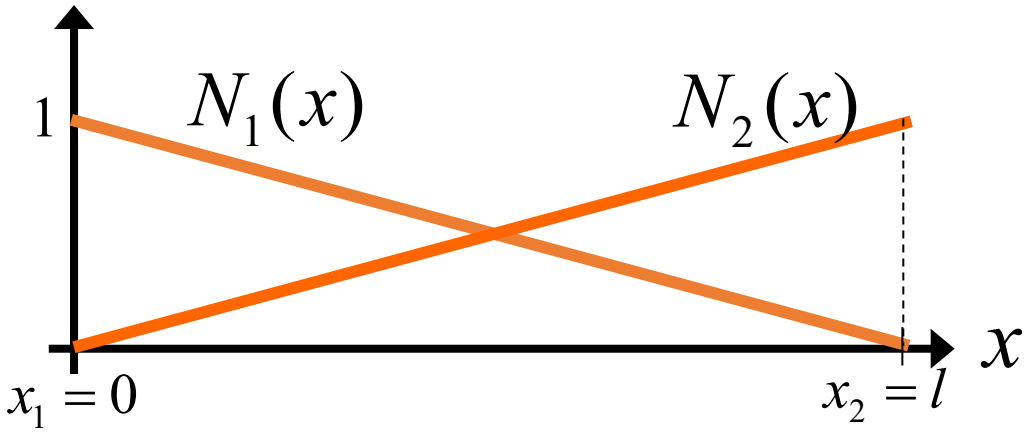


$$N_{n+1}(x) = \frac{x - x_n}{h_n} \qquad x_n \le x \le x_{n+1}$$

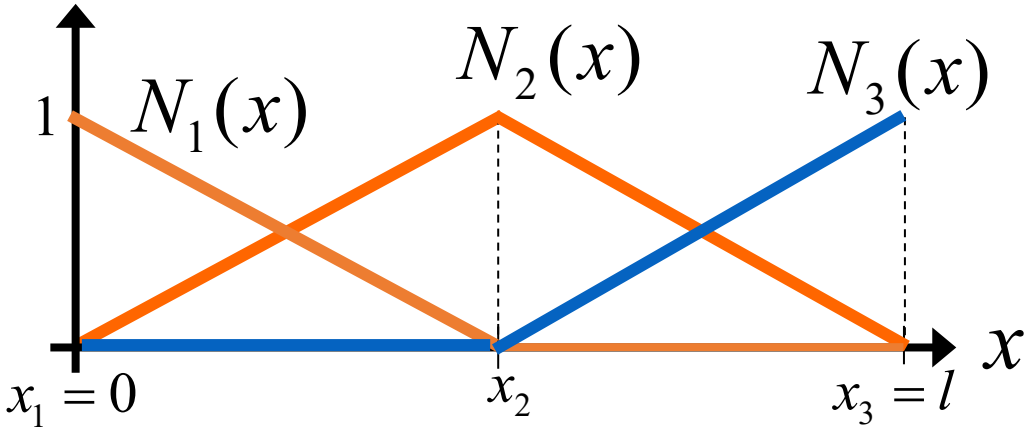


Global shape functions for a different number of elements

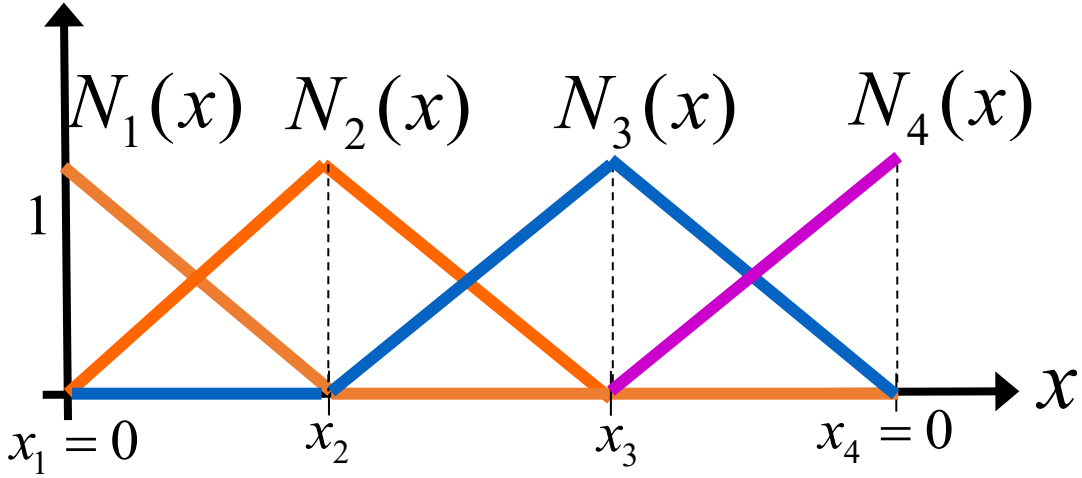
$n = 1$



$n = 2$



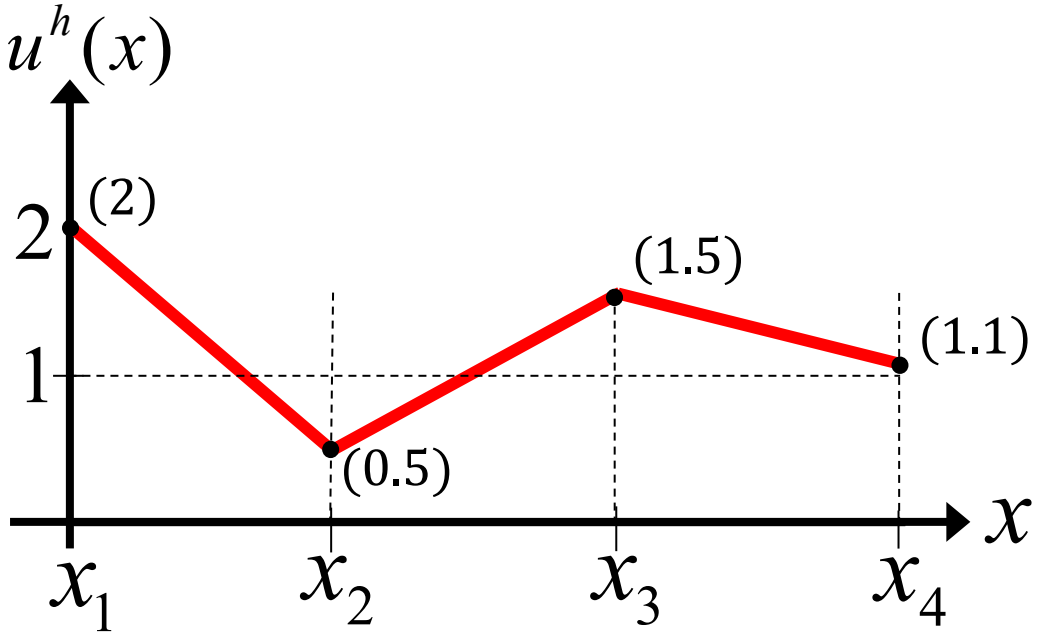
$n = 3$

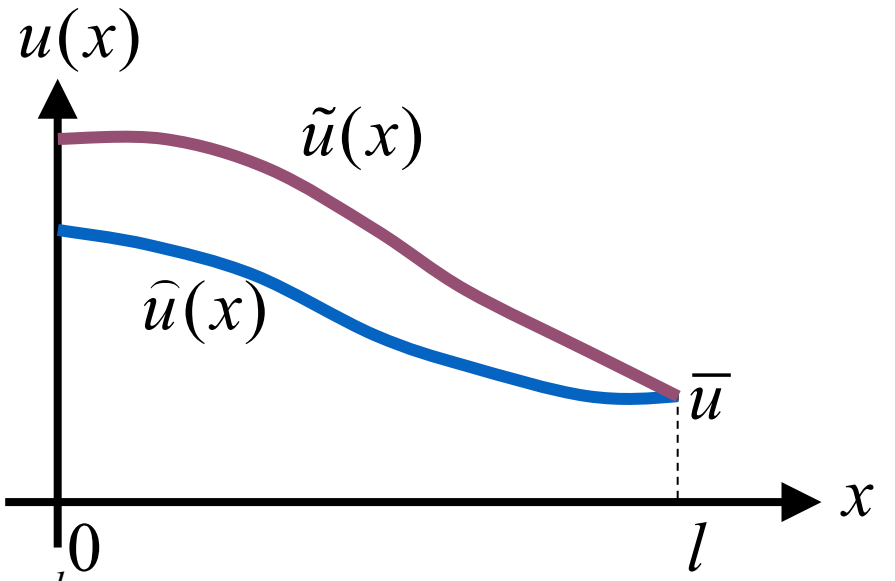


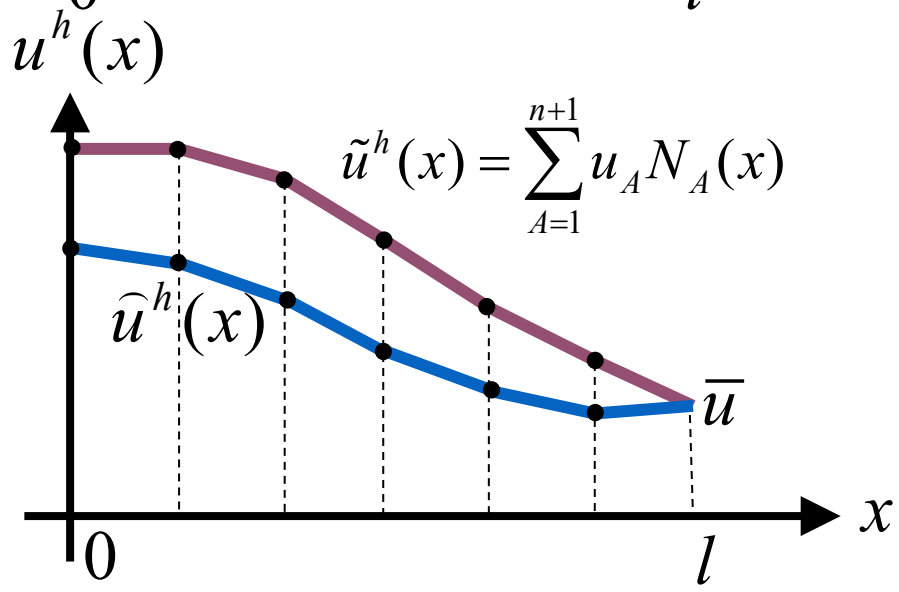
Approximation of trial functions

The key point is to observe how a continuous **trial solution** can be approximated in terms of **global shape functions**

$$u^h(x) = 2N_1(x) + 0.5N_2(x) + 1.5N_3(x) + 1.1N_4(x)$$







Approximation of trial functions (cont.)

- In the general case the approximated function $u^h(x)$ is written as

$$u^h(x) = \sum_{A=1}^{n+1} u_A N_A(x)$$

- Note that the shape function satisfies the following property

$$N_A(x_B) = \delta_{AB} = \begin{cases} 1 & \text{si } A = B \\ 0 & \text{other case} \end{cases}$$

- Function $N_A(x)$ is different from zero only in the elements containing x_A

- The test function is approximated in a similar way:

$$w^h(x) = \sum_{A=1}^{n} c_A N_A(x)$$

Discretization of the weak form

Weak form of the second differential equation

$$\int_0^l \frac{\partial w}{\partial x}\frac{\partial u}{\partial x}dx = -w(0)\bar{v} + \int_0^l wf dx \qquad \forall w \,/\, w(l) = 0$$

The objective is to replace w and u by their approximations w^h and u^h

$$w^h(x) = \sum_{A=1}^{n} c_A N_A(x) \qquad u^h(x) = \sum_{B=1}^{n} u_B N_B(x) + \bar{u} N_{n+1}(x)$$

Approximation of the first derivative

$$w^h(x) = \sum_{A=1}^{n} c_A N_A(x)$$

$$\frac{\partial w(x)}{\partial x} \cong \frac{\partial w^h(x)}{\partial x} = \sum_{A=1}^{n} c_A \frac{\partial N_A(x)}{\partial x}$$

$$u^h(x) = \sum_{B=1}^{n} u_B N_B(x) + \overline{u}\, N_{n+1}(x)$$

$$\frac{\partial u(x)}{\partial x} \cong \frac{\partial u^h(x)}{\partial x} = \sum_{B=1}^{n} u_B \frac{\partial N_B(x)}{\partial x} + \overline{u}\frac{\partial N_{n+1}(x)}{\partial x}$$

Approximation of the left hand side

$$\int_0^l \frac{\partial w}{\partial x}\frac{\partial u}{\partial x}dx \cong \int_0^l \frac{\partial w^h}{\partial x}\frac{\partial u^h}{\partial x}dx$$

$$= \int_0^l \left(\sum_{A=1}^{n} c_A \frac{\partial N_A(x)}{\partial x}\right)\left(\sum_{B=1}^{n} u_B \frac{\partial N_B(x)}{\partial x} + \overline{u}\frac{\partial N_{n+1}(x)}{\partial x}\right)dx$$

$$= \sum_{A=1}^{n} c_A \sum_{B=1}^{n} u_B \left(\int_0^l \frac{\partial N_A}{\partial x}\frac{\partial N_B}{\partial x}dx\right) + \sum_{A=1}^{n} c_A \left(\int_0^l \frac{\partial N_A}{\partial x}\frac{\partial N_{n+1}}{\partial x}dx\right)\overline{u}$$

Approximation of the right hand side

$$\int_0^l f(x)w(x)dx \cong \int_0^l f(x)\left(\sum_{A=1}^{n} c_A N_A(x)\right)dx$$

$$= \sum_{A=1}^{n} c_A \left(\int_0^l f(x)N_A(x)dx\right)$$

Approximation of the third term

$$\bar{v}w(0) = \bar{v}\sum_{A=1}^{n} c_A N_A(0)$$

$$= \sum_{A=1}^{n} c_A \bar{v}N_A(0)$$

Computation of the integrals

$$k_{AB} = \int_0^l \frac{\partial N_A}{\partial x}\frac{\partial N_B}{\partial x}dx$$

$$f_A = \int_0^l f(x)N_A(x)dx$$

$$g_A = \int_0^l \frac{\partial N_A}{\partial x}\frac{\partial N_{n+1}}{\partial x}dx$$

$$v_A = \bar{v}N_A(0)$$

$$k_{AB} = \int_0^l \frac{\partial N_A}{\partial x}\frac{\partial N_B}{\partial x}dx$$

$$\mathbf{K} = \begin{bmatrix} \int_0^l \frac{\partial N_1}{\partial x}\frac{\partial N_1}{\partial x}dx & \cdots & \int_0^l \frac{\partial N_1}{\partial x}\frac{\partial N_n}{\partial x}dx \\ \vdots & & \vdots \\ \int_0^l \frac{\partial N_n}{\partial x}\frac{\partial N_1}{\partial x}dx & \cdots & \int_0^l \frac{\partial N_n}{\partial x}\frac{\partial N_n}{\partial x}dx \end{bmatrix}$$

$$f_A = \int_0^l f(x)N_A(x)dx$$

$$\mathbf{f} = \begin{bmatrix} \int_0^l f(x)N_1(x)dx \\ \vdots \\ \int_0^l f(x)N_n(x)dx \end{bmatrix}$$

$$g_A = \int_0^l \frac{\partial N_A}{\partial x}\frac{\partial N_{n+1}}{\partial x}dx$$

$$\mathbf{g} = \begin{bmatrix} \int_0^l \frac{\partial N_1}{\partial x}\frac{\partial N_{n+1}}{\partial x}dx \\ \vdots \\ \int_0^l \frac{\partial N_n}{\partial x}\frac{\partial N_{n+1}}{\partial x}dx \end{bmatrix}$$

Formulation of the global system

$$\int_0^l \frac{\partial w^h}{\partial x}\frac{\partial u^h}{\partial x}dx = -w^h(0)\bar{v} + \int_0^l w^h f dx \qquad \forall w^h \;/\; w^h(l) = 0$$

A) $$\int_0^l \frac{\partial w^h}{\partial x}\frac{\partial u^h}{\partial x}dx = \sum_{A=1}^{n} c_A \sum_{B=1}^{n} k_{AB}u_B + \sum_{A=1}^{n} c_A g_A \bar{u} = \mathbf{c}^T\mathbf{K}\mathbf{d} + \bar{u}\mathbf{c}^T\mathbf{g}$$

B) $$\int_0^l w^h f dx = \sum_{A=1}^{n} c_A f_A = \mathbf{c}^T\mathbf{f}$$

C) $$\bar{v}w^h(0) = \bar{v}c_1 = \mathbf{c}^T\bar{\mathbf{v}}$$

Global system of equations

$$\int_0^l \frac{\partial w^h}{\partial x}\frac{\partial u^h}{\partial x}dx = -w^h(0)\bar{v} + \int_0^l w^h f dx \qquad \forall w^h \;/\; w^h(l) = 0$$

$$\mathbf{c}^T\mathbf{K}\mathbf{d} \;=\; \mathbf{c}^T\bar{\mathbf{v}} \;+\; \mathbf{c}^T\mathbf{f} \qquad \forall \mathbf{c}^T = \begin{pmatrix} c_1 & \cdots & c_n \end{pmatrix}$$

rearranging and factoring the coefficients we can obtain the following equation system:

$$\mathbf{c}^T\left(\mathbf{K}\mathbf{d} + \bar{u}\mathbf{g} - \bar{\mathbf{v}} - \mathbf{f}\right) = 0 \qquad \forall \mathbf{c}^T = \begin{pmatrix} c_1 & \cdots & c_n \end{pmatrix}$$

$$\boxed{\mathbf{K}\mathbf{d} = \mathbf{F}^{ext}} \qquad \mathbf{F}^{ext} = \mathbf{f} + \bar{\mathbf{v}} - \bar{u}\mathbf{g}$$

Summary

Find $u(x)$ such that

$$\frac{\partial^2 u}{\partial x^2} + f(x) = 0 \qquad 0 < x < l$$

$$u(l) = \overline{u}$$

$$\frac{\partial u}{\partial x}(0) = \overline{v}$$

Find $u(x)$ continuous function that satisfies

$$u(l) = \overline{u}$$

$$\int_0^l \frac{\partial w}{\partial x}\frac{\partial u}{\partial x}dx = -w(0)\overline{v} + \int_0^l wf dx \qquad \forall w \,/\, w(l) = 0$$

$$\int_0^l \frac{\partial w^h}{\partial x}\frac{\partial u^h}{\partial x}dx = -w^h(0)\overline{u} + \int_0^l w^h f dx \qquad \forall w^h \,/\, w^h(l) = 0$$

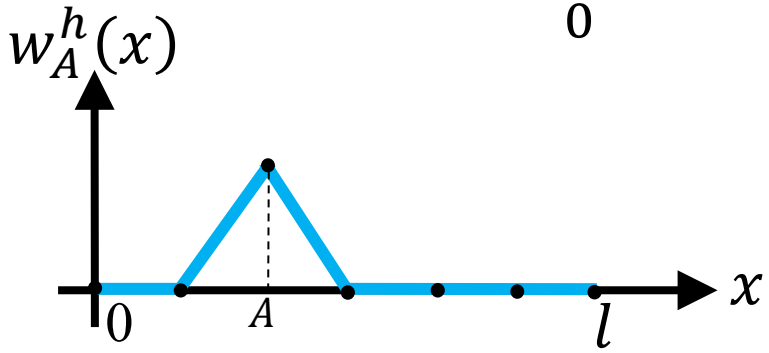
$$\mathbf{Kd} = \mathbf{F}^{ext}$$

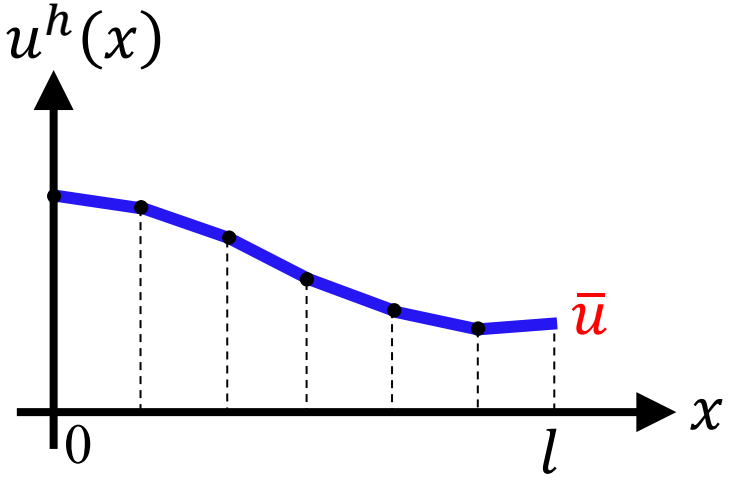
APPENDIX A

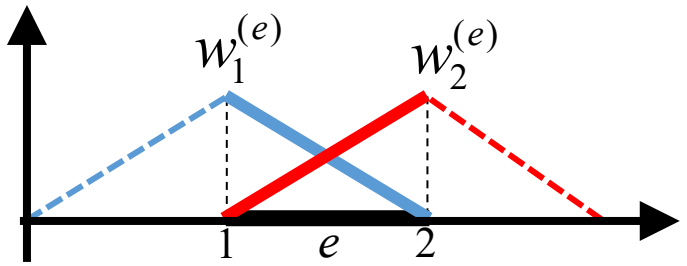
CONSTRUCTION OF ELEMENT STIFFNESS MATRICES

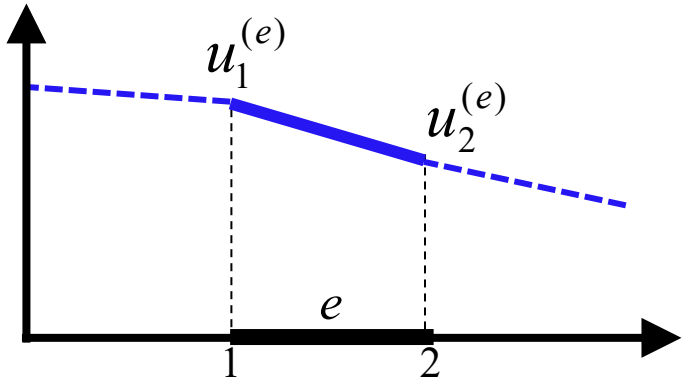
Global integration vs element integration

$$\int_0^l \frac{\partial w^h}{\partial x}\frac{\partial u^h}{\partial x}dx = \sum_{e=1}^{Nel}\int_{x_1^e}^{x_2^e} \frac{\partial w_e^h}{\partial x}\frac{\partial u_e^h}{\partial x}dx$$

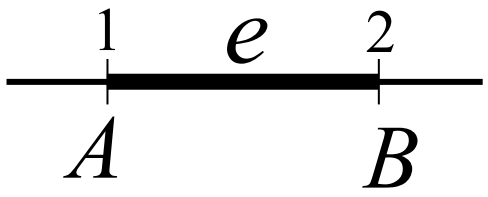


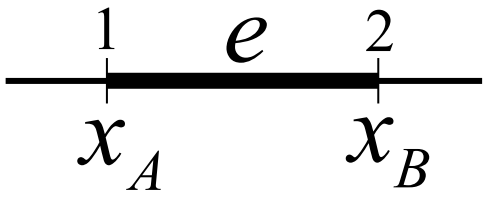


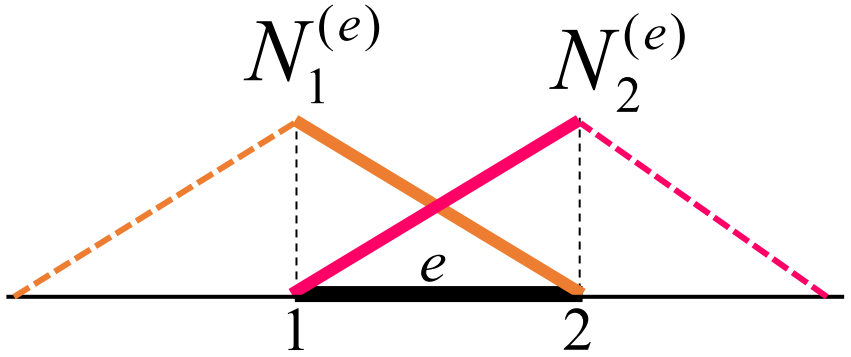




Element functions







Shape functions

$$N_1^e(x) = \frac{1}{h^e}\left(x_B - x\right)$$

$$N_2^e(x) = \frac{1}{h^e}\left(x - x_A\right)$$

$$h^e = \left|x_B - x_A\right|$$

Element trial function

$$u^e(x) = u_1^e N_1^e(x) + u_2^e N_2^e(x) = \begin{pmatrix} N_1^e & N_2^e \end{pmatrix} \begin{pmatrix} u_1^e \\ u_2^e \end{pmatrix} = \mathbf{N}^e \mathbf{u}^e$$

$$w^e(x) = c_1^e N_1^e(x) + c_2^e N_2^e(x) = \begin{pmatrix} N_1^e & N_2^e \end{pmatrix} \begin{pmatrix} c_1^e \\ c_2^e \end{pmatrix} = \mathbf{N}^e \mathbf{c}^e$$

Element stiffness matrix

Derivative of the trial function

$$\frac{\partial u^e(x)}{\partial x} = u_1^e \frac{\partial N_1^e}{\partial x} + u_2^e \frac{\partial N_2^e}{\partial x} = \underbrace{\begin{pmatrix} \frac{\partial N_1^e}{\partial x} & \frac{\partial N_2^e}{\partial x} \end{pmatrix}}_{\mathbf{B}^e} \underbrace{\begin{pmatrix} u_1^e \\ u_2^e \end{pmatrix}}_{\mathbf{u}^e} = \mathbf{B}^e \mathbf{u}^e$$

The term of the left

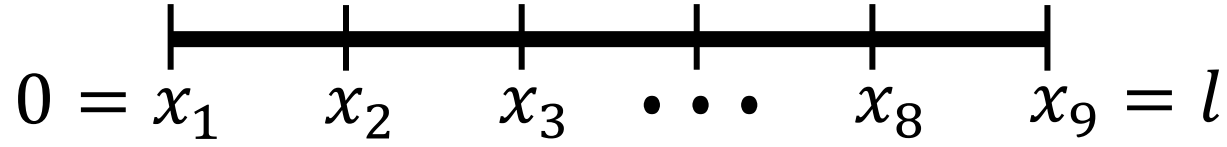
$$\int_{x_1^e}^{x_2^e} \frac{\partial w^e}{\partial x} \frac{\partial u^e}{\partial x} dx = \int_{x_1^e}^{x_2^e} \mathbf{c}^{e^T} \mathbf{B}^{e^T} \mathbf{B}^e \mathbf{u}^e dx = \mathbf{c}^{e^T} \underbrace{\left(\int_{x_1^e}^{x_2^e} \mathbf{B}^{e^T} \mathbf{B}^e dx \right)}_{\mathbf{K}^e} \mathbf{u}^e$$

Element stiffness matrix

$$\mathbf{K}^e = \int_{\Omega_e} \mathbf{B}^{e^T} \mathbf{B}^e dx = \frac{1}{h^e} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$$

Global matrix representation

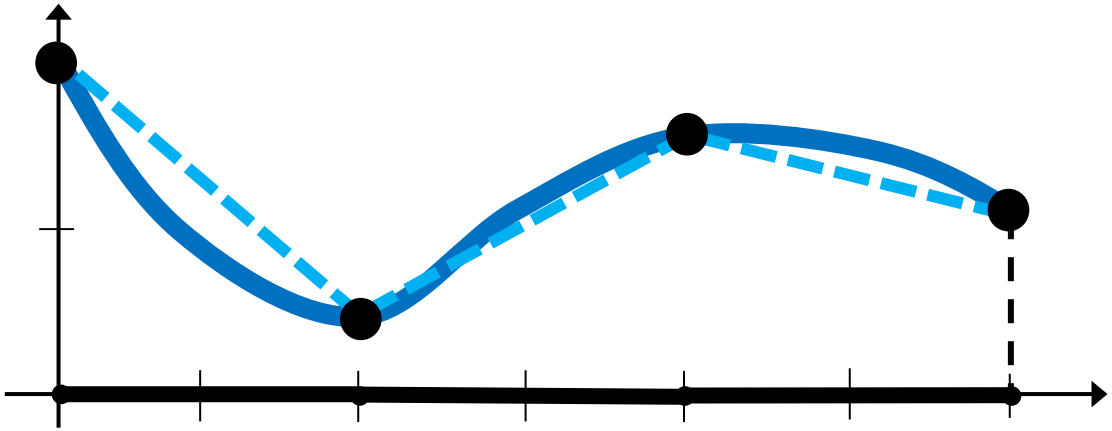
The stiffness matrix is a banded matrix in which the bandwidth is quite small compared with the total number of degrees of freedom.



$$\mathbf{K} = \begin{bmatrix} k_{11} & k_{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ k_{21} & k_{22} & k_{23} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & k_{32} & k_{33} & k_{34} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & k_{43} & k_{44} & k_{45} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & k_{54} & k_{55} & k_{56} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & k_{65} & k_{66} & k_{67} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & k_{76} & k_{77} & k_{78} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & k_{87} & k_{88} & k_{89} \\ & & & & & & & k_{98} & k_{99} \end{bmatrix}$$

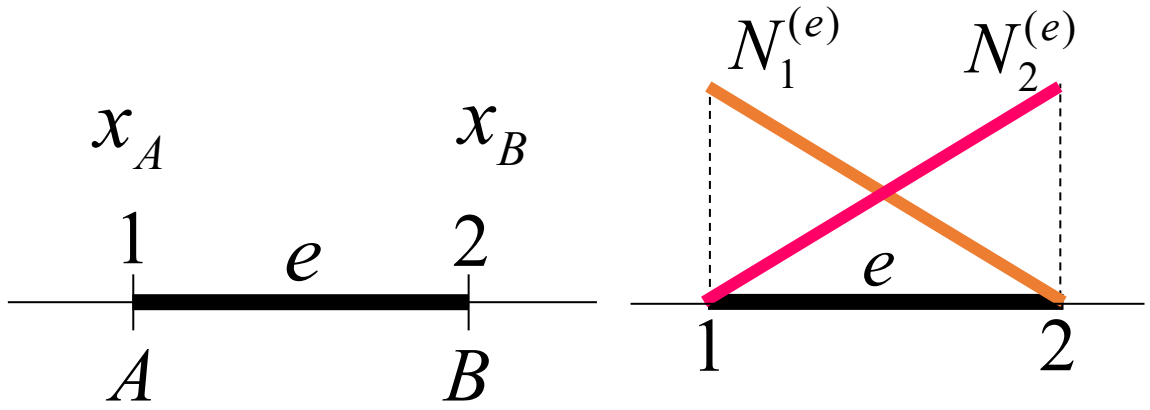
APPENDIX B

Higher order of approximation at element level



Quadratic order of approximation

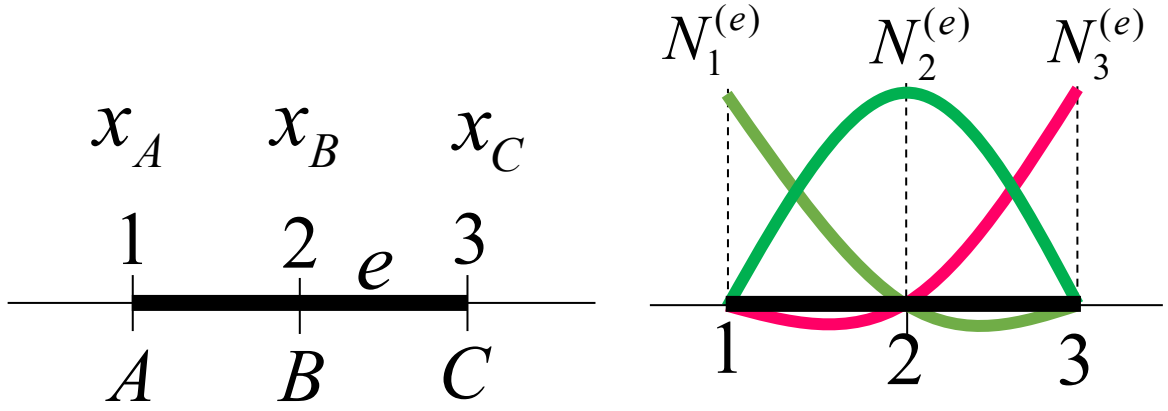
Linear approximation



$$u^e = u_1^e N_1^e + u_2^e N_2^e$$

$$\frac{\partial u^e}{\partial x} = u_1^e \frac{\partial N_1^e}{\partial x} + u_2^e \frac{\partial N_2^e}{\partial x}$$

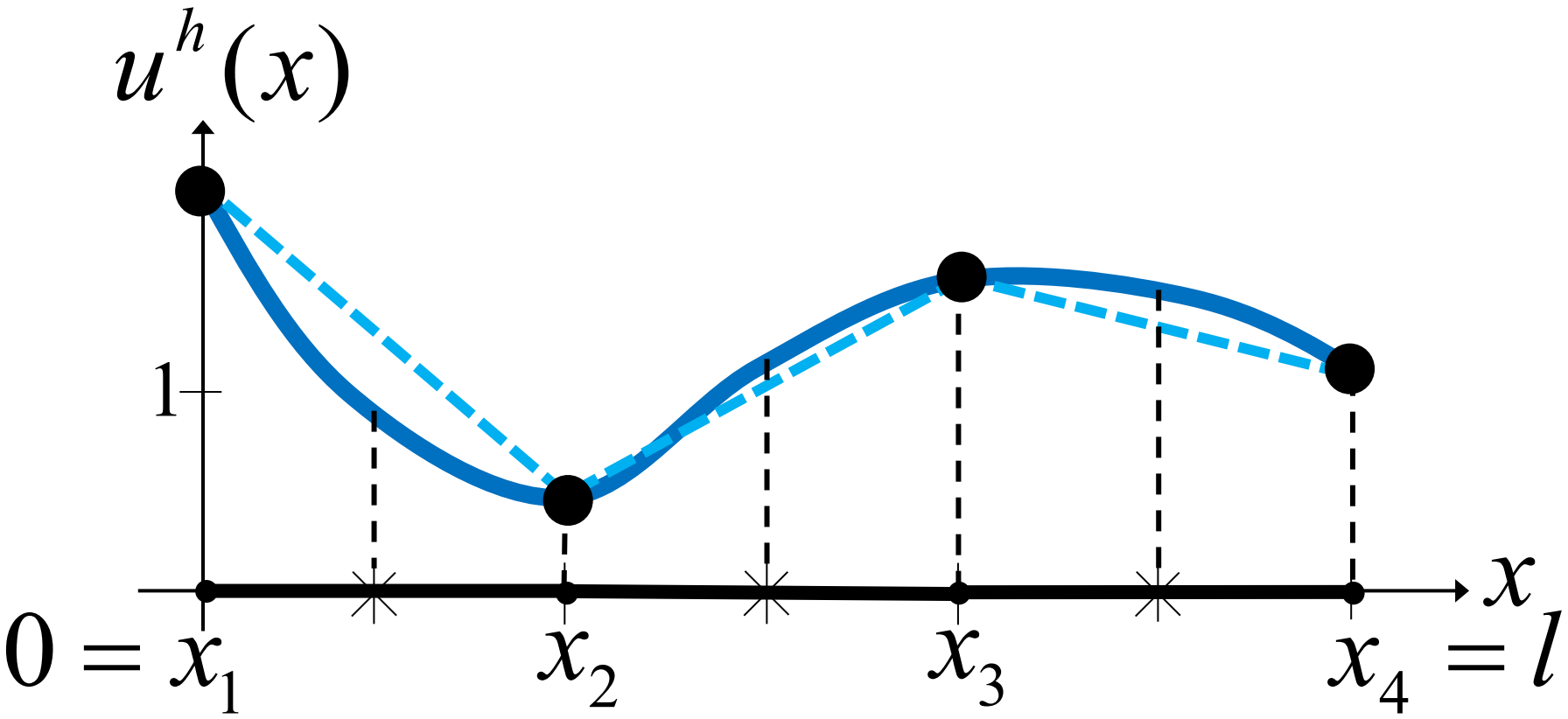
Quadratic approximation



$$u^e = u_1^e N_1^e + u_2^e N_2^e + u_3^e N_3^e$$

$$\frac{\partial u^e}{\partial x} = u_1^e \frac{\partial N_1^e}{\partial x} + u_2^e \frac{\partial N_2^e}{\partial x} + u_3^e \frac{\partial N_3^e}{\partial x}$$

Improvement in the approximation of the trial function



New derivative of the trial function

$$\frac{\partial u^e(x)}{\partial x} = u_1^e \frac{\partial N_1^e}{\partial x} + u_2^e \frac{\partial N_2^e}{\partial x} + u_3^e \frac{\partial N_3^e}{\partial x} = \underbrace{\begin{pmatrix} \frac{\partial N_1^e}{\partial x} & \frac{\partial N_2^e}{\partial x} & \frac{\partial N_3^e}{\partial x} \end{pmatrix}}_{\mathbf{B}^e} \underbrace{\begin{pmatrix} u_1^e \\ u_2^e \\ u_3^e \end{pmatrix}}_{\mathbf{u}^e} = \mathbf{B}^e \mathbf{u}^e$$

New element internal work

$$\int_{\Omega_e} \frac{\partial w^e}{\partial x} \frac{\partial u^e}{\partial x} dx = \int_{\Omega_e} \mathbf{c}^{e^T} \mathbf{B}^{e^T} \mathbf{B}^e \mathbf{u}^e dx = \mathbf{c}^{e^T} \underbrace{\left(\int_{\Omega_e} \mathbf{B}^{e^T} \mathbf{B}^e dx \right)}_{\mathbf{K}^e} \mathbf{u}^e$$

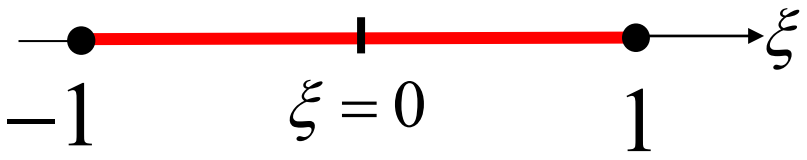
Element stiffness matrix component

$$\mathbf{K}_{ij}^e = \int_{x_1^e}^{x_2^e} \frac{\partial N_i^e}{\partial x} \frac{\partial N_j^e}{\partial x} dx$$

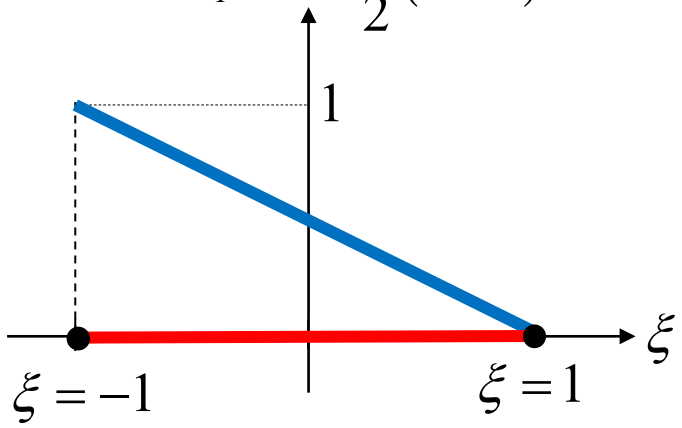
APPENDIX C

Definition of the standard element

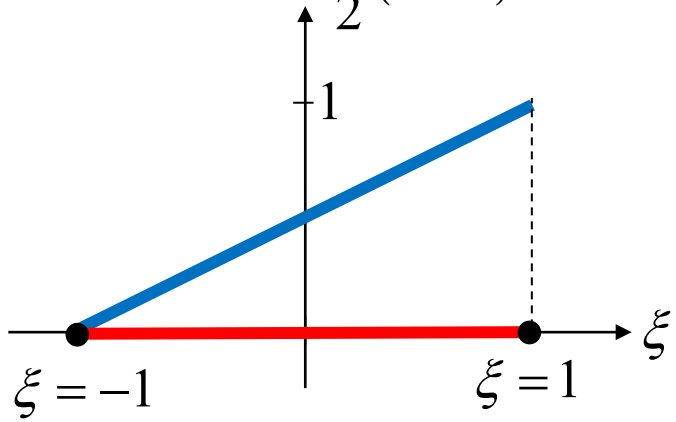
Standard element and linear shape functions



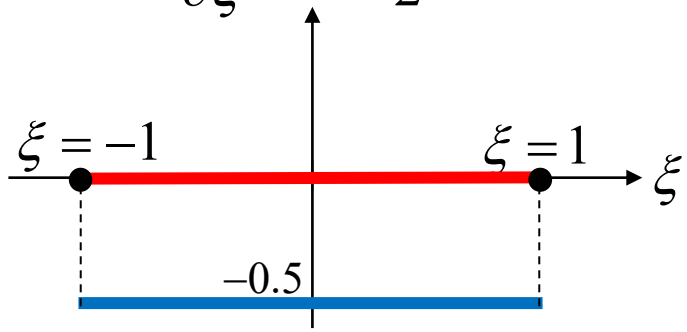
$$\hat{N}_1(\xi) = \frac{1}{2}(1-\xi)$$



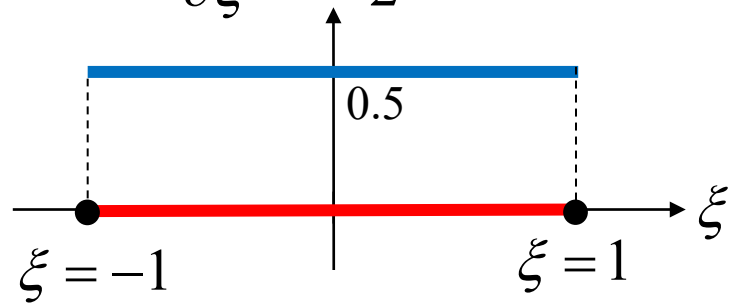
$$\hat{N}_2(\xi) = \frac{1}{2}(1+\xi)$$



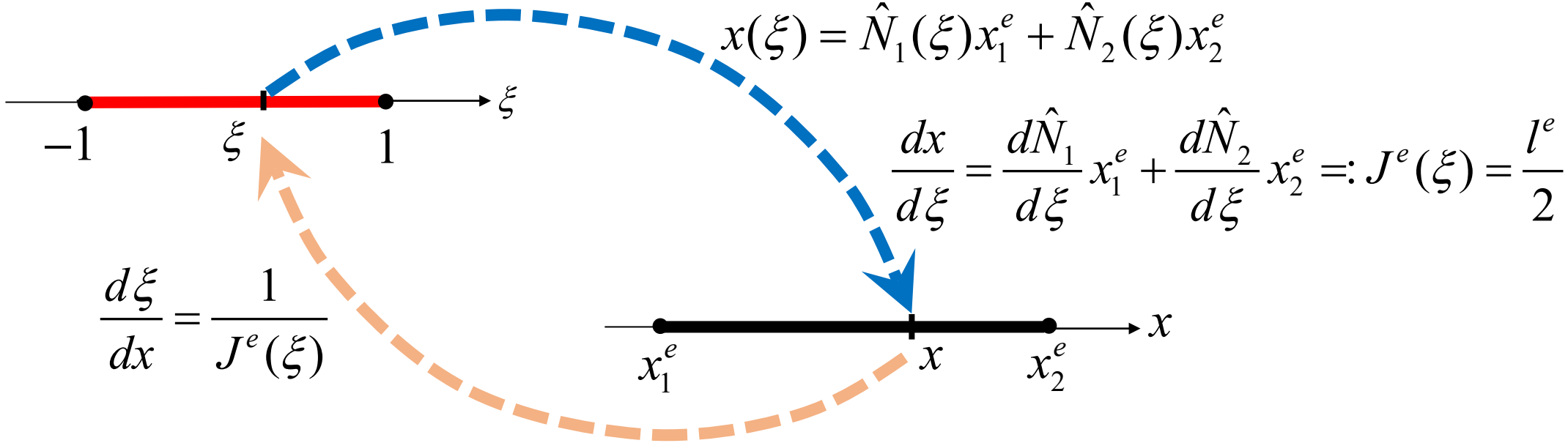
$$\frac{\partial \hat{N}_1(\xi)}{\partial \xi} = -\frac{1}{2}$$



$$\frac{\partial \hat{N}_2(\xi)}{\partial \xi} = \frac{1}{2}$$



Linear shape functions defined on an arbitrary element



$$N_1(x) := \hat{N}_1(\xi(x))$$

$$\frac{\partial N_1(x)}{\partial x} = \frac{1}{J^e(\xi)}\frac{\partial \hat{N}_1}{\partial \xi}$$

$$N_2(x) := \hat{N}_2(\xi(x))$$

$$\frac{\partial N_2(x)}{\partial x} = \frac{1}{J^e(\xi)}\frac{\partial \hat{N}_2}{\partial \xi}$$

Comput. of the stiffness matrix in terms of the standard element

$$\mathbf{K}_{ij}^{e} = \int_{x_1^e}^{x_2^e} \frac{\partial N_i^e}{\partial x} \frac{\partial N_j^e}{\partial x} dx$$

$$dx = J^e(\xi) d\xi$$

$$\int_{x_1^e}^{x_2^e} (\bullet)\, dx = \int_{-1}^{1} (\bullet)\, J^e(\xi) d\xi$$

$$\frac{\partial N_i(x)}{\partial x} = \frac{1}{J^e(\xi)} \frac{\partial \hat{N}_i}{\partial \xi}$$

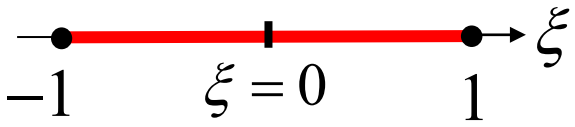
$$\frac{\partial N_j(x)}{\partial x} = \frac{1}{J^e(\xi)} \frac{\partial \hat{N}_j}{\partial \xi}$$

$$\frac{\partial N_i^e}{\partial x} \frac{\partial N_j^e}{\partial x} = \frac{1}{J^e(\xi)^2} \frac{\partial \hat{N}_i}{\partial \xi} \frac{\partial \hat{N}_j}{\partial \xi}$$

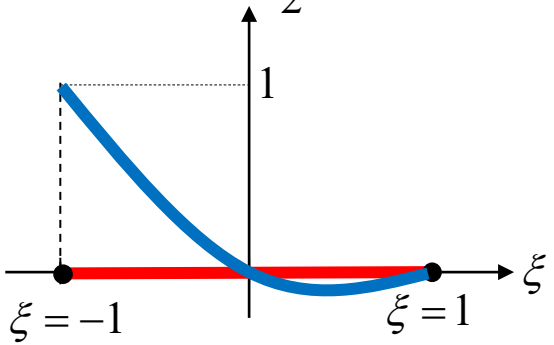
$$\mathbf{K}_{ij}^{e} = \int_{-1}^{1} \frac{1}{J^e(\xi)} \frac{\partial \hat{N}_i}{\partial \xi} \frac{\partial \hat{N}_j}{\partial \xi} d\xi$$

Standard element and **quadratic shape functions**

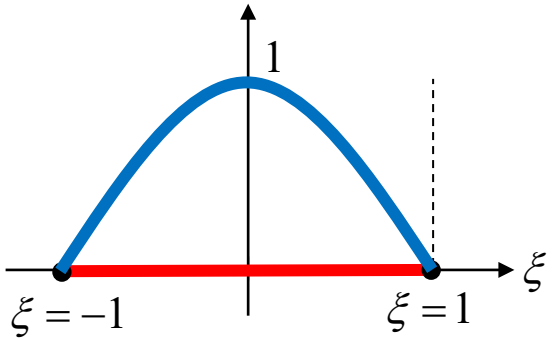
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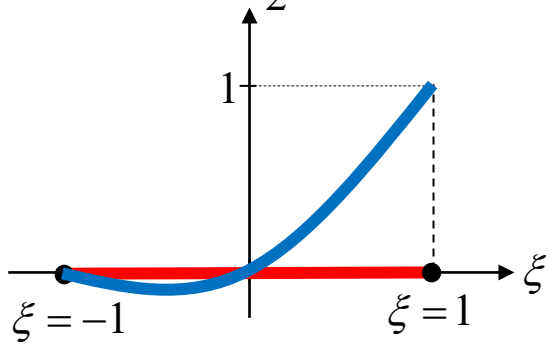
$$\hat{N}_1(\xi) = \frac{1}{2}\xi(\xi - 1)$$



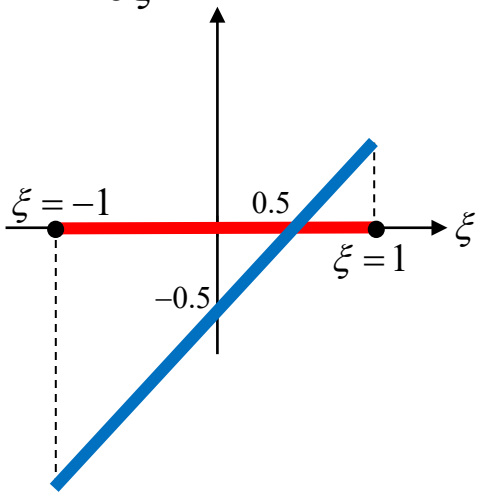
$$\hat{N}_2(\xi) = \left(1 - \xi^2\right)$$



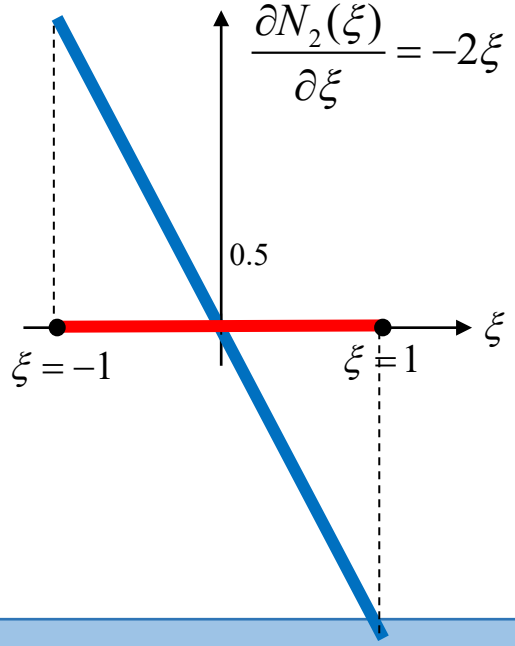
$$\hat{N}_3(\xi) = \frac{1}{2}\xi\left(\xi + 1\right)$$



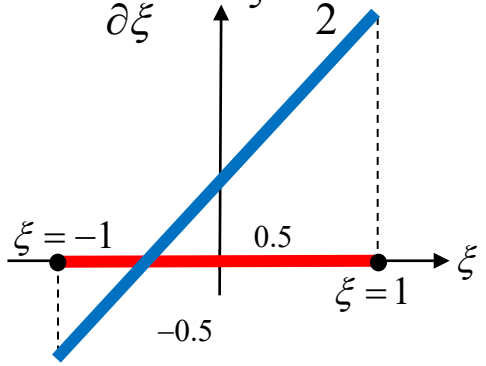
$$\frac{\partial \hat{N}_1(\xi)}{\partial \xi} = \xi - \frac{1}{2}$$



$$\frac{\partial \hat{N}_2(\xi)}{\partial \xi} = -2\xi$$

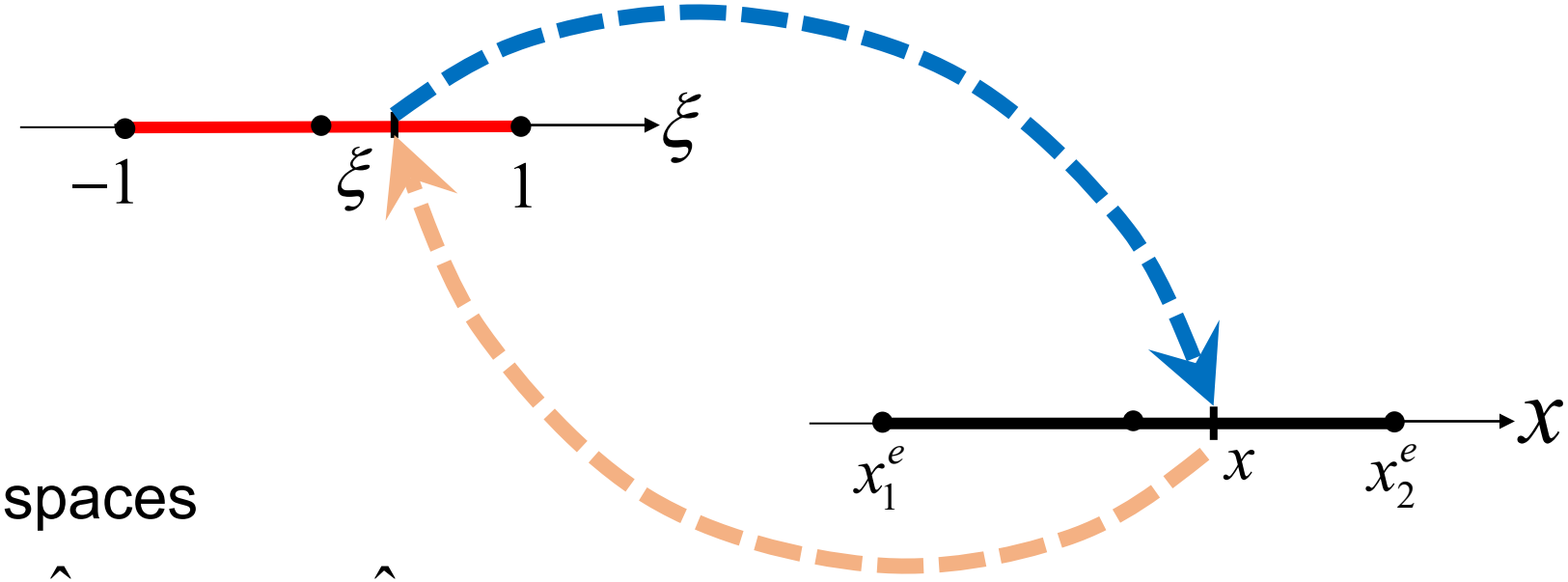


$$\frac{\partial \hat{N}_3(\xi)}{\partial \xi} = \xi + \frac{1}{2}$$



Quadratic shape funct. defined on an arbitrary element

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Transformation between spaces

$$x(\xi) = \hat{N}_1(\xi)x_1^e + \hat{N}_2(\xi)x_2^e + \hat{N}_3(\xi)x_3^e$$

Jacobian of the transformation

$$J^e(\xi) = \frac{dx(\xi)}{d\xi} = \frac{d\hat{N}_1}{d\xi}x_1^e + \frac{d\hat{N}_2}{d\xi}x_2^e + \frac{d\hat{N}_3}{d\xi}x_3^e$$

Definition of the cartesian shape function

$$N_i(x(\xi)) := \hat{N}_i(\xi)$$

Computation of the cartesian derivative

$$\frac{\partial N_i(x(\xi))}{\partial \xi} = \frac{\partial \hat{N}_i(\xi)}{\partial \xi}$$

$$\frac{\partial N_i(x(\xi))}{\partial \xi} = \frac{\partial N_i(x)}{\partial x}\frac{\partial x(\xi)}{\partial \xi} = \frac{\partial N_i(x)}{\partial x} J^e(\xi)$$

$$\frac{\partial N_i(x)}{\partial x} = \left(J^e(\xi)\right)^{-1}\frac{\partial \hat{N}_i(\xi)}{\partial \xi}$$

Component of the stiffness matrix

$$\mathbf{K}_{ij}^e = \int_{-1}^{1} \left(J^e(\xi)\right)^{-1} \frac{\partial \hat{N}_i}{\partial \xi}\frac{\partial \hat{N}_j}{\partial \xi} d\xi$$

APPENDIX C

Numerical Integration

Gauss Quadrature

- **In general**, the integral representing the weak form cannot be integrated in closed form.
- There are many **numerical techniques** !!
- **Gauss Quadrature** is one of the most efficient for functions that are **polynomials** or **nearly polynomials**.
- Our objective is to compute the following integral:

$$\int_{-1}^{1} f(\xi)d\xi = ?$$

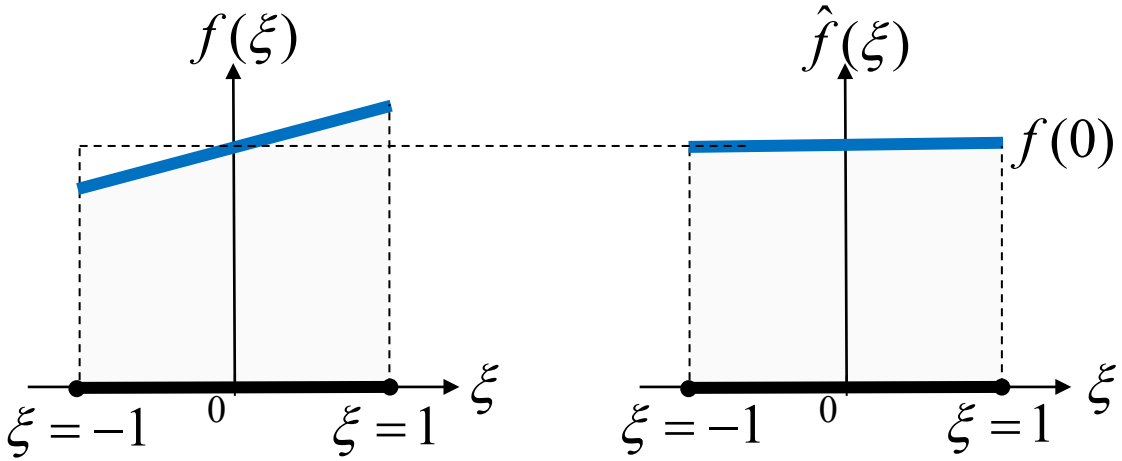
- In the Gauss integration procedure, we approximate the integral by the following expression

$$\int_{-1}^{1} f(\xi)d\xi =$$

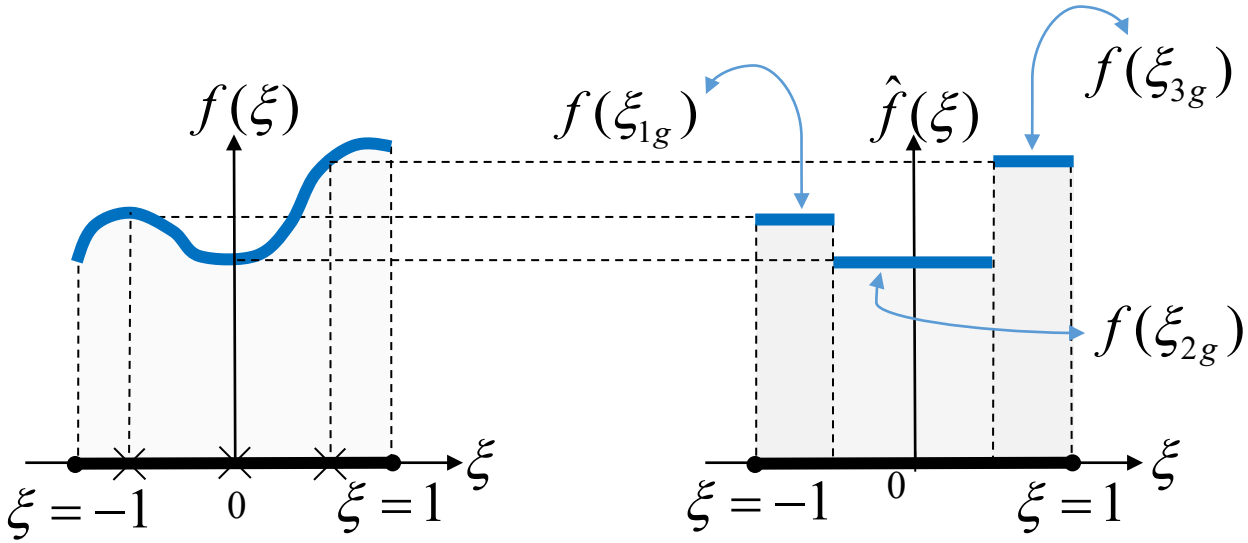
$$w_1 f(\xi_1) + w_1 f(\xi_2) + \cdots + w_{ng} f(\xi_{ng})$$

$$\int_{-1}^{1} f(\xi)d\xi = \sum_{kg=1}^{ng} w_{kg} f(\xi_{kg})$$

Remark



$$\int_{-1}^{1} f(\xi)d\xi = w_{1g} f(\xi_{1g}) = \underbrace{2}_{w_{1g}} \underbrace{f(0)}_{f(\xi_{1g})}$$



$$\int_{-1}^{1} f(\xi)d\xi \cong w_{1g} f(\xi_{1g}) + w_{2g} f(\xi_{2g}) + w_{3g} f(\xi_{3g})$$

Gauss Quadrature problem

The idea of the Gauss integration is to choose the weights and the integration points so that a given polynomial is integrated exactly !

$$p(\xi) = c_0 + c_1\xi + \cdots + c_n\xi^n$$

$$\int_{-1}^{1} p(\xi)d\xi = c_0\xi + \frac{1}{2}c_1\xi^2 + \cdots + \frac{1}{n+1}c_n\xi^{n+1}\Big|_{-1}^{1} = \hat{I}$$

The problem

The problem is to **find the number** ng the **weights** $\{w_{kg}\}$ and **quadrature points** $\{\xi_{kg}\}$ such that the quadrature formula gives the exact integral for the polynomial $p(\xi)$

$$\sum_{kg=1}^{ng} w_{kg}\, p(\xi_{kg}) = \hat{I}$$

It is a system of **nonlinear algebraic equations** for the unknown weights and points.

Weights and quadrature points

$$ng \geq \frac{n+1}{2}$$

ng	ξ_i	w_i
1	0.0	2.0
2	$\pm 1/\sqrt{3}$	1.0
3	±0.7745966692	0.555 555 5556
	0.0	0.888 888 8889
4	±0.8611363116	0.347 854 8451
	±0.3399810436	0.652 145 1549
5	±0.9061798459	0.236 926 8851
	±0.5384693101	0.478 628 6705
	0.0	0.568 888 8889
6	±0.9324695142	0.171 324 4924
	±0.6612093865	0.360 761 5730
	±0.2386191861	0.467 913 9346

APPENDIX C

Convergence of the FEM

- In the assessment of solution quality for various types of elements or various degrees of discretization, a measure of element performance is clearly required.
- For these purposes, an exact solution is needed (In many cases, that exact solution is constructed by 'manufacturing' the solution)
- How can the error in a finite element solution $u^h(x)$ be quantified if we know the exact solution, $u^{ex}(x)$?

Norm of functions

$$\mathbf{a} \in \mathbb{R}^n$$

$$\|\mathbf{a}\| = \left(\sum_{i=1}^{n} a_i^2 \right)^{\frac{1}{2}}$$

$$f : [x_1, x_2] \rightarrow \mathbb{R}$$

$$\|f\|_{L_2} = \left(\int_{x_1}^{x_2} f^2(x) dx \right)^{\frac{1}{2}}$$

The error in a finite element solution is defined by

$$\|e\|_{L_2} = \left\|u^{ex}(x) - u^h(x)\right\|_{L_2} = \left(\int_0^L \left(u^{ex}(x) - u^h(x)\right)^2 dx\right)^{\frac{1}{2}}$$

where $u^{ex}(x)$ is the exact solution and $u^h(x)$ is the finite element solution.

In comparing errors of different solutions, it is recommendable to normalize the error by the norm of the exact solution. The normalized error is given by

$$\|e\|_{L_2} = \frac{\left\|u^{ex}(x) - u^h(x)\right\|_{L_2}}{\left\|u^{ex}(x)\right\|_{L_2}} = \frac{\left(\int_0^L \left(u^{ex}(x) - u^h(x)\right)^2 dx\right)^{\frac{1}{2}}}{\left(\int_0^L \left(u^{ex}(x)\right)^2 dx\right)^{\frac{1}{2}}}$$

Remark

- The integrals are computed by subdividing the domain into elements.
- The typical procedure is to use Gauss quadrature in each element.
- Usually, higher order Gauss quadrature formulas a required because the exact solution is generally not a polynomial.

Computation of the error

$$I^{res} = \sum_{e=1}^{n_{el}} \left(\sum_{kg=1}^{ng} J^e(\xi_{kg}) \left(r^e(\xi_{kg}) \right)^2 w_{kg} \right)$$

$$I^{ex} = \sum_{e=1}^{n_{el}} \left(\sum_{kg=1}^{ng} J^e(\xi_{kg}) \left(u^{ex}(x^e(\xi_{kg})) \right)^2 w_{kg} \right)$$

$$e^u = \frac{I^{res}}{I^{ex}}$$

where

$$x^e(\xi_{kg}) = \sum_{i=1}^{n_{ne}} \hat{N}_i(\xi_{kg}) x_i^e$$

$$u^{h,e}(\xi_{kg}) = \sum_{i=1}^{n_{ne}} \hat{N}_i(\xi_{kg}) \hat{u}_i^e$$

$$J^e(\xi_{kg}) = \sum_{i=1}^{n_{ne}} \frac{d\hat{N}_i(\xi_{kg})}{d\xi} x_i^e$$

$$r^e(\xi_{kg}) = u^{ex}(x^e(\xi_{kg})) - u^{h,e}(\xi_{kg})$$

APPENDIX D

Time integration method

Newmark Approximations

$$\dot{d}_{n+1} = \frac{\gamma}{\beta\Delta t}\left(d_{n+1} - d_n\right) - \left(\frac{\gamma}{\beta} - 1\right)\dot{d}_n - \left(\frac{\gamma}{2\beta} - 1\right)\Delta t\ddot{d}_n$$

$$\ddot{d}_{n+1} = \frac{1}{\beta\Delta t^2}\left(d_{n+1} - d_n\right) - \frac{1}{\beta\Delta t}\dot{d}_n - \left(\frac{1}{2\beta} - 1\right)\ddot{d}_n$$

$$\gamma = \frac{1}{2}, \quad \beta = \frac{1}{4}$$